

Problem 7.12

A store is running a promotion during which customers have two options for payment. Option one is to pay 90% of the purchase price two months after the date of sale. Option two is to deduct $X\%$ off the purchase price and pay cash on the date of sale. A customer wishes to determine X such that he is indifferent between the two options when valuing them using an effective annual interest rate of 8%. Which of the following equations would the customer need to solve?

(A) $\left(\frac{X}{100}\right)\left(1 + \frac{0.08}{6}\right) = 0.90$

(B) $\left(1 - \frac{X}{100}\right)\left(1 + \frac{0.08}{6}\right) = 0.90$

(C) $\left(\frac{X}{100}\right)(1.08)^{1/6} = 0.90$

(D) $\left(\frac{X}{100}\right)\left(\frac{1.08}{1.06}\right) = 0.90$

(E) $\left(1 - \frac{X}{100}\right)(1.08)^{1/6} = 0.90$

Solution

Let the purchase price be P .

Under option 1, the customer pays $0.9P$ two months after the sale date. Since two months is $\frac{1}{6}$ of a year, the present value at the date of sale is

$$\frac{0.9P}{(1.08)^{1/6}}.$$

Under option 2, the customer pays cash immediately after deducting $X\%$ from the purchase price, so the amount paid at the date of sale is

$$P\left(1 - \frac{X}{100}\right).$$

For the customer to be indifferent between the two options, these values must be equal:

$$P\left(1 - \frac{X}{100}\right) = \frac{0.9P}{(1.08)^{1/6}}.$$

Cancelling P and multiplying both sides by $(1.08)^{1/6}$ gives

$$\left(1 - \frac{X}{100}\right)(1.08)^{1/6} = 0.90.$$

Therefore, the correct equation is

$$(E) \quad \left(1 - \frac{X}{100}\right) (1.08)^{1/6} = 0.90.$$